Arunabha Majumder

Wearable Mechanical Engineer

Aalborg, Denmark | +45 71 62 24 00
somrkmv1997@gmail.com | 97arunabhasit027@gmail.com
Portfolio: arunabha22.github.io/arunabha-majumder
LinkedIn: linkedin.com/in/arunabha-majumder-681264107



PROFILE

Mechanical engineer who designs hardware worn on the human arm. Since 2018 I have worked on an arm exoskeleton and exo glove, pneumatic muscles for a soft exosuit and a shoulder exoskeleton. I take wearable parts from sketches and SolidWorks CAD to 3D printed prototypes, test them on real people and iterate.

EXPERIENCE

PhD Researcher, Robotics (project VIEXO)
Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton worn on the arm and shoulder: a variable stiffness mechanism with a worm gear for pretension, working in parallel with a 6 Nm motor. It uses more than 25% less motor energy, which means longer battery life (Best Paper Award, IFToMM ISRM 2026).
- Built the CAD assembly and drawings in SolidWorks, optimised the mechanism with a genetic algorithm, 3D printed parts for fit testing, refined them and then had the final parts manufactured.
- Integrated the motor, encoder and torque/force sensors; measured the stiffness on a physical test setup.
- Collected data from 12 participants for a biomechanics study, recording EMG with the Qualisys system.
- Design for manufacture: replaced a clutch that kept breaking and was hard to make with a simple bolt-locking clutch (8 locking angles) that was easy to make, fit and replace.
- Designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM); estimated its stiffness with less than 10% error and quantified human arm impedance (6 participants, Qualisys motion capture).
- Taught university courses and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Soft Exosuit (Pneumatic Muscles)
CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic artificial muscles for a soft exosuit; designed the outer PID loop of a neural network-based controller in MATLAB/Simulink and implemented it in LabVIEW (IEEE CONECCT 2022).

Bachelor's Capstone and Student Teams
Siddaganga Institute of Technology, India | *2017 - 2019*

- Designed and built an arm exoskeleton and an exo glove for a paralysed arm: CAD, machining, assembly and testing (TATA Technologies Prayatna Innovation Fund grant; Best Major Project Award).
- Built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018) with student teams.

SKILLS

CAD: SolidWorks (CSWA): parametric parts and assemblies, technical drawings, tolerances, fasteners, clearances, organised assembly files; Fusion 360 (basic, last used 2019)

Wearables and prototyping: arm exoskeletons, exo glove, soft exosuit (pneumatic muscles), testing with participants, 3D printing (Bambu Lab, Ultimaker Cura), hands-on assembly, DFM and DFA

Human testing: Qualisys motion capture and EMG data collection, torque/force sensors

Integration: motors, encoders, Arduino, MATLAB/Simulink, LabVIEW

Art and graphic design: hand sketching of concepts, painting, technical illustration and visuals for papers, posters and demos; Adobe Illustrator, Photoshop, Procreate

EDUCATION

PhD, Mechatronics and Robotics, Aalborg University, Denmark | 2023 - present (research complete)

MTech, Mechatronics, AcSIR - CSIR-CMERI, India | 2020 - 2022

BTech, Mechanical Engineering, Siddaganga Institute of Technology, India | 2015 - 2019

Senior Diploma in Fine Arts (Painting), Bangiya Sangeet Parishad | 2008, First Division with Distinction

LANGUAGES AND REFERENCES

English (C1), Bengali (native), Hindi (spoken)

References: Prof. Shaoping Bai (sbh@mp.aau.dk); Assoc. Prof. Lars Rosgaard Jensen; Assoc. Prof. Anderson de Souza Castelo Oliveira; all Department of Materials and Production, Aalborg University